

John Whitton

Principal Software Engineer — Trading Infrastructure — Execution Systems — Distributed Systems
Rust | Market Infrastructure | DeFi Protocol Engineering
Mountain View, CA — john@johnwhitton.com — (415) 336-6194
johnwhitton.com — portfolio.johnwhitton.com — jincubator.com — linkedin.com/in/johnwhitton

Profile

Principal software engineer and technical leader specializing in Rust-based trading infrastructure, execution systems, distributed systems, and financial infrastructure.

Over the past eighteen months, I deliberately focused more deeply on hands-on engineering in trading systems, Rust, DeFi, and distributed infrastructure, designing and implementing production-oriented systems directly. This builds on more than two decades of hands-on engineering, architecture, and technical leadership across blockchain infrastructure, payments, enterprise systems, and distributed platforms.

I combine current implementation experience with system-level architecture and technical leadership, with a particular focus on performance, correctness, reliability, and operational clarity.

Engineering Highlights

- Designed and implemented Salus, a Rust/Tokio trading and distributed-systems platform spanning real-time market ingestion, graph-based route evaluation, simulation, execution, replay, and strategy analysis.
- Built and measured a retained Ethereum topology containing 3,415 tokens, 4,552 pool components, and 2,401,108 catalog routes; retained Base workloads measured 16,537 sustained and 38,462 peak route evaluations/sec.
- Engineered bounded asynchronous processing, deterministic execution boundaries, exact arithmetic, state-freshness controls, execution telemetry, and multi-chain processing across EVM networks.
- Consulted on DeFi infrastructure spanning leverage tokens, auction-based rebalancing, Morpho lending integrations, CoW Protocol execution, ERC-6909 accounting, and protocol invariants.
- Led engineering for stablecoin payment infrastructure, cross-chain settlement, routing, proving, wallets, and Layer-2 transaction execution at Eco.
- Designed and implemented Kanga Finance, a Solidity-based DEX spanning AMMs, liquidity, staking, incentives, yield strategies, flash loans, and protocol administration.
- Designed and implemented EAVE, a Rust/Substrate Polkadot parachain and DeFi platform spanning runtime modules, staking, liquidity, EVM compatibility, APIs, and cross-chain architecture.
- Led engineering teams and technically complex initiatives, including an 8-engineer Eco team and Harmony engineering growth from 6 to 20 engineers.

Representative Engineering Work

Salus (Rust)	Designed and implemented a Rust/Tokio trading platform spanning real-time market ingestion, graph construction, affected-route selection, simulation, execution, replay, and strategy analysis. Retained Ethereum topology contained 2.4M+ catalog routes; bounded asynchronous pipelines, deterministic execution boundaries, exact arithmetic, freshness controls, and execution evidence support block-sensitive multi-chain processing.	2025–Present
Seamless (Solidity/Rust)	Solidity and Rust components for leverage tokens, auction-based rebalancing, Morpho lending integrations, CoW Protocol requests, ERC-6909 accounting, and protocol invariant analysis.	2025–2026
Eco (Solidity/TypeScript)	Stablecoin payment infrastructure, programmable money, cross-chain settlement, routing, proving, wallet services, and Layer-2 transaction execution.	2023–2025
1wallet (Solidity)	One-time-password smart contract wallet eliminating seed phrase management through cryptographic OTP authentication.	2022–2023
1ns (Go/Solidity)	ENS-style decentralized naming service for Harmony using Go and Solidity.	2022–2023
Kanga Finance (Solidity)	Designed and implemented Solidity DEX infrastructure spanning AMM pools, multi-asset liquidity, staking, protocol incentives, yield strategies, flash loans, fee conversion, and liquidity migration.	2022–2023
EAVE (Rust)	Designed and built a Rust/Substrate Polkadot parachain and DeFi platform spanning runtime modules, consensus integration, governance, staking, liquidity, EVM compatibility, APIs, and cross-chain architecture; deployed standalone testnet and parachain environments.	2020–2022
Harmony (Go/Solidity)	Layer-1 blockchain infrastructure including mainnet launch, validator onboarding, staking, SDKs, Ledger integration, APIs, and ecosystem tooling.	2019–2020
Loyyal (Go)	Enterprise blockchain, transaction-processing APIs, tokenization, mobile applications, and partner integration platforms.	2016–2018
Enterprise Integration	Service-oriented platforms integrating SAP, Salesforce, payments, ecommerce, licensing, logistics, and partner ecosystems.	1999–2016

Core Technologies

Languages	Rust, Solidity, Go, TypeScript, Python
Trading Infrastructure	Market data, execution, graph-based routing, simulation, replay, strategy evaluation, MEV, DeFi
Distributed Systems	Rust, Tokio, async/concurrency, streaming, bounded queues, graph processing, deterministic execution, service-oriented architecture
Data Platforms	PostgreSQL, ClickHouse (evaluation), MongoDB
Cloud & DevOps	AWS, Docker, Kubernetes, Terraform, CI/CD
Observability	Datadog, PagerDuty
Blockchain / Runtime	Ethereum/EVM, Base, Arbitrum, Unichain, Optimism, Polkadot, Substrate, FRAME, Harmony, Hyperledger
DeFi / Protocol	AMMs, liquidity pools, staking, lending, flash loans, intent execution, MEV
Developer Tooling	Tycho, Alloy, Substreams, Foundry
AI Engineering	AI-assisted software engineering, architecture and documentation automation, code review, Codex, ChatGPT, GitHub Copilot, Claude

Technical Portfolio

Selected technical evidence covering trading-system architecture, distributed systems, Rust implementation, protocol engineering, architecture decisions, and source-backed code walkthroughs. Detailed material is available for technical review.

<https://portfolio.johnwhitton.com>

Career Experience

Principal Engineer / Founder

2018 – Present

Jincubator

- Since 2025, deliberately focused more deeply on hands-on engineering, designing and implementing Rust trading infrastructure, DeFi protocols, distributed systems, and supporting technical research.
- Designed and implemented Salus across market ingestion, graph-based evaluation, simulation, execution, replay, strategy analysis, telemetry, and deterministic evidence boundaries.
- Consulted on DeFi leverage-token infrastructure, auction-based rebalancing, lending integrations, intent execution, and protocol invariants.
- Built and contributed to blockchain and protocol systems including 1wallet, 1ns, Kanga Finance, EAVE, solver/intent infrastructure, and supporting engineering tooling.
- Applied AI-assisted engineering workflows across implementation, architecture, testing, documentation, and code review while retaining deterministic engineering and validation boundaries.

Engineering Manager / Senior Protocol Engineer

Aug 2023 – Jan 2025

Eco Inc

- Led a team of 8 engineers building stablecoin payment infrastructure, programmable money systems, routing, proving, wallet, and protocol services.
- Delivered cross-chain routing, settlement, and transaction execution infrastructure across Ethereum and Layer-2 ecosystems while recruiting engineers and managing delivery.

Ecosystem Architect

Aug 2019 – May 2020

Harmony Protocol

- Supported Harmony Mainnet launch, staking infrastructure, APIs, SDKs, validator tooling, and ecosystem integrations.
- Coordinated 50+ production network upgrades, deployed 300+ nodes, supported 600+ validators, and scaled engineering from 6 to 20 engineers.

Chief Technology Officer

Jul 2016 – Jun 2018

Loyyal

- Led distributed teams building enterprise blockchain, loyalty, transaction-processing APIs, Hyperledger, private Ethereum, and partner integration platforms.

Earlier Experience

Enterprise Architect / Engineering Manager / Founder — EFI, Autodesk, Global Enterprise Solutions

- Built enterprise SOA platforms spanning SAP, Salesforce, licensing, ecommerce, payments, logistics, and partner systems.
- Led architecture across 100+ enterprise initiatives, managed \$3M+ planning portfolios, and founded a consulting firm generating approximately \$800K annual revenue.

Education

Bachelor of Computer Science, University of Queensland